

HPC Trig-Model Project
Victoria Harbour Over 2 Days
2024-2025

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HPC Trig Model Project

Introduction:

Victoria Harbour is a natural landform harbour in Hong Kong that separates Hong Kong Island in the south from the Kowloon Peninsula. Understanding and modeling tidal patterns is essential in as tides impact marine navigation, coastal construction, and ecosystems. Tides are primarily driven by gravitational interactions between the Earth, moon, and sun, resulting periodic fluctuations in sea levels. These fluctuations can be accurately represented using trigonometric functions, which this paper will focus on.

This paper will attempt to analyze the tidal data for Victoria Harbour over a 43-hour period, from July 1st to July 2nd, 2024. Using collected data points for tide heights at regular intervals, a sinusoidal function will be constructed to model the tidal behavior observed in this period. By fitting a trigonometric function to the data, we aim to provide insights into the harbor's tidal patterns and demonstrate the applicability of trigonometric models in tide waves.



Variable Definitions:

t : time in hours from July 1st to 2nd, 2024

y : water height in meters

R^2 : coefficient of determination

$h_1(t)$: hand calculated graph/equation, shown in Figure 2

$h_2(t)$: Desmos generated graph, graph of Figure 3

a, b, h, k : variables for transformations (scale factor, translations) in sinusoidal equations

** These are the variables that are used in this paper, refer back to this section.*

We are given data of the total cases for the two days (43-hour period). It will be presented in a table, showcasing the dates and time of the day in hours, as well as showing the water height in meters.

The data of total cases for the two days is given in Table 1 below ...

Table 1: Tidal Data for Victoria Harbour from July 1st to 2nd, 2024 ($t = 0$, midnight)

Date			Water Height in Meters		
7/1/2024 1:00	1	16.8	7/1/2024 23:00	23	18.5
7/1/2024 2:00	2	13.4	7/2/2024 0:00	24	20.4
7/1/2024 3:00	3	10.5	7/2/2024 1:00	25	21.7
7/1/2024 4:00	4	8.9	7/2/2024 2:00	26	22.5
7/1/2024 5:00	5	9	7/2/2024 3:00	27	22.5
7/1/2024 6:00	6	11.1	7/2/2024 4:00	28	21.4
7/1/2024 7:00	7	14.7	7/2/2024 5:00	29	19.2
7/1/2024 8:00	8	18.7	7/2/2024 6:00	30	15.9
7/1/2024 9:00	9	22.1	7/2/2024 7:00	31	12.1
7/1/2024 10:00	10	24.2	7/2/2024 8:00	32	8.9
7/1/2024 11:00	11	24.9	7/2/2024 9:00	33	7.2
7/1/2024 12:00	12	24.1	7/2/2024 10:00	34	7.6
7/1/2024 13:00	13	22.3	7/2/2024 11:00	35	10.2
7/1/2024 14:00	14	19.5	7/2/2024 12:00	36	14.3
7/1/2024 15:00	15	15.9	7/2/2024 13:00	37	18.5
7/1/2024 16:00	16	12.6	7/2/2024 14:00	38	21.7
7/1/2024 17:00	17	10.5	7/2/2024 15:00	39	23.2
7/1/2024 18:00	18	9.7	7/2/2024 16:00	40	23.1
7/1/2024 19:00	19	10.1	7/2/2024 17:00	41	21.8
7/1/2024 20:00	20	11.6	7/2/2024 18:00	42	19.7
7/1/2024 21:00	21	13.8	7/2/2024 19:00	43	17.3
7/1/2024 22:00	22	16.3			

**The measurement of tide waves began on July 1st, at 1:00AM. "1" represents one hour since 12:00AM on July 1, 2024.*

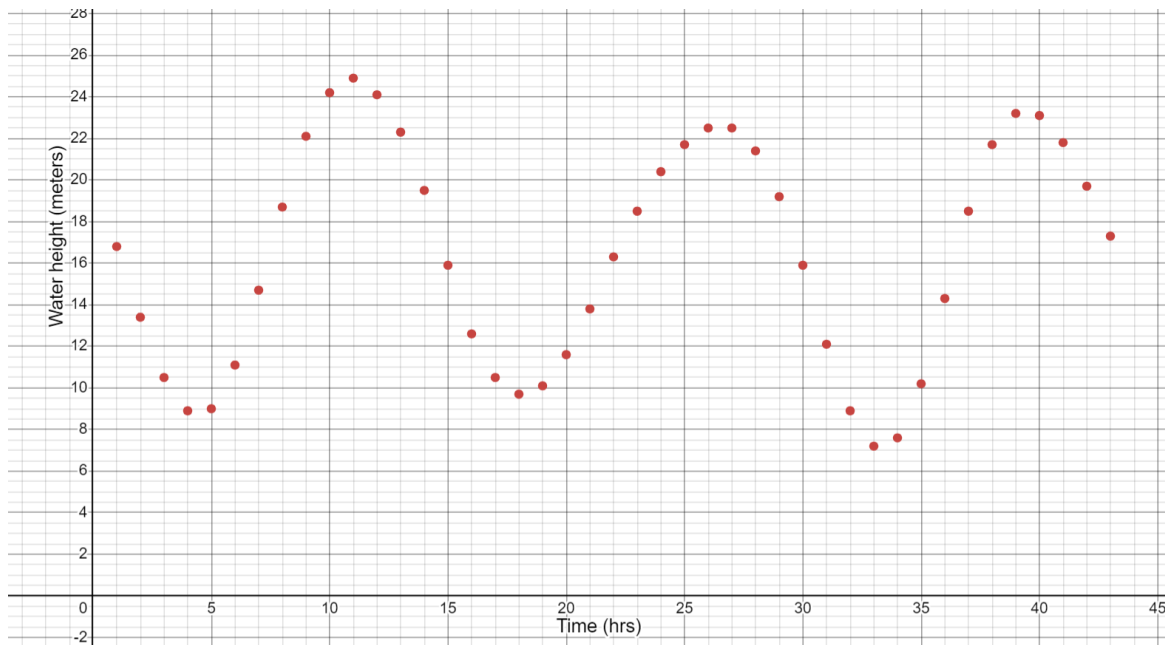
The measurements on the table highlight the fluctuations of water height over time per hour over a 43-hour period.

After showing these measurements on the table, we will now create a scatterplot of the data given in Table 1 to get a visual representation of the pattern. We will paste the data into Desmos to get a technologically accurate scatterplot, and it will be shown in Figure 1 below with red scatterplot points.

Based on the data, we expect to see a sinusoidal wave with different amplitudes and periods across the scatterplot. We expect for the minimum point on the scatterplot to be 7.20, and the maximum point to be 24.9.

Figure 1 will be plotted by Desmos and will be shown below:

Figure 1: Scatterplot of Data



**Graph of data points given, plotted in a scatterplot*

The x axis is labeled as the time (t) in hours and y axis is labeled in the water height in meters to give a visual representation of the fluctuations of water height in Victoria Harbour.

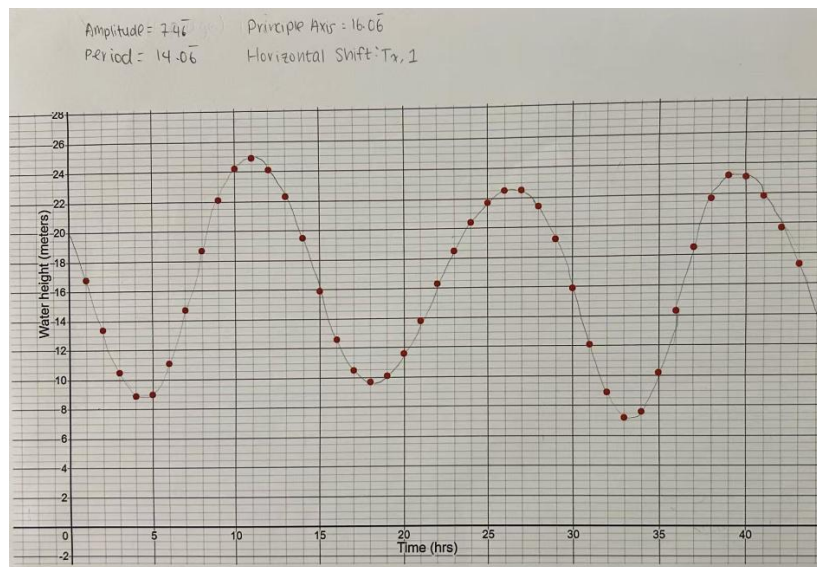
The scatterplot in Figure 1 shows the sinusoidal behavior that we expected. The pattern does not fit any parent function that we are familiar with, therefore, to find the line of best fit, we will take the average values of each period (amplitude, principle axis, transformations in the t and y axis).

However, the shape of the scatterplot represents the shape of a transformed sine graph with different periods and amplitudes for each wavelength. We decided and therefore concluded that a sine graph would be the model that would best fit the data showcased in the scatterplot given above in Figure 1. While looking at technologically simulated graph dotted on Desmos, we saw a total of almost 3 periods as the 3rd period cut off midway on the graph.

Ultimately, we concluded that it was enough data to create a valid model, as we decided to take the average amplitudes of the periods to craft a proper sinusoidal equation that would fit the plotted points the best.

We will now draw the curve of best fit drawn by hand and provide the manually calculated values of our sine graph below...

Figure 2: Curve of Best Fit Drawn by Hand



**Note that this is not the line of best fit, it is just the data points linked together. The line of best fit is given below:*

$$h_1(t) = -7.46\sin(0.45(t-1)) + 16.1$$

Calculation of Amplitude:

To find a mathematical model, we decided to begin by finding each variable within the equation, $y_1 \sim a \sin(b(t_1 - h)) - k$. In order to find the variable a , or the amplitude, we first found all the maximums of each period we saw on the scatterplot, which were 24.9, 22.5, and 23.2. Then we found the minimums, which were 8.90, 9.70, and 7.20. After that, we calculated the average of the maximums by adding them together and then dividing by three.

$$\frac{(24.9 + 22.5 + 23.2)}{3} = 23.5\bar{3} \approx 23.5$$

Next, we repeated the same process and calculated the average of the minimums by adding them together and then dividing by three.

$$\frac{(8.9 + 9.7 + 7.2)}{3} = 8.60$$

Then, we subtracted the average of the minimums from the average of the maximums. This gave us the distance between the average of the maximums and the average of the minimums, which we divided by two to find the average amplitude.

$$23.5\bar{3} - 8.6 = 14.9\bar{3} \approx 14.9$$

$$\frac{14.9\bar{3}}{2} = 7.4\bar{6} \approx 7.47$$

Once we found that the amplitude was approximately equal to 7.47, we started to solve for the principal axis for our mathematical model.

Calculation of the Principal Axis:

In order to find the variable k , or the principal axis, we took the average of the maximums and the average of the minimums that we found in the calculations above and added them together. After that, we divided the sum by two to find a line equally distanced in between the maximum and the minimum, or the principal axis.

$$23.5\bar{3} + 8.6 = 32.1\bar{3}$$

$$\frac{32.1\bar{3}}{2} = 16.0\bar{6} \approx 16.1$$

Once we found that the principal axis was approximately equal to 16.1, we began to solve for the variable b in our mathematical model.

Calculation of Variable b :

To find variable b , we figured that we first needed to solve for the period, because the period is equal to $\frac{2\pi}{b}$. To solve for the period, we decided to calculate each of the three periods

individually, and then solve for the average. In order to calculate each period, we analyzed the scatterplot, and mentally drew a curve that would connect the points closest to the principal axis. Then, we imagined where the curve would cross the principal axis to find the start of the period, and then repeated the same process for the end of the period. Afterwards, we subtracted the end of the period from the start of the period, giving us an estimated calculation of one of the periods. When solving for the end of one period, the result would also represent the start of the next period, so we continued until we found all three periods. Finally, we added them all together and divided by three to find the average of the three periods.

For the first period, we found that the start was $(1.2, 16.0\bar{6})$ and the end was $(14.96, 16.0\bar{6})$. To find the period, we simply found the difference between the t values.

$$14.96 - 1.2 = 13.76 \approx 13.8$$

For the second period, we found that the start was $(14.96, 16.0\bar{6})$ and the end was $(29.96, 16.0\bar{6})$. To find the period, we repeated the same process used above.

$$29.96 - 14.96 = 15.0$$

Finally, for the third period, we found that the start was $(29.96, 16.0\bar{6})$ and the end was $(43.4, 16.0\bar{6})$. After that, we repeated the same process to find the third and final period.

$$43.4 - 29.96 = 13.44$$

Next, we calculated the sum of all three periods and divided by three. This gave us the average of the periods.

$$\frac{13.76 + 15.0 + 13.44}{3} = 14.0\bar{6} \approx 14.1$$

For the last step, we solved for the variable b , by using the equation: $period = \frac{2\pi}{b}$.

$$14.0\bar{6} = \frac{2\pi}{b}$$

$$b = \frac{2\pi}{14.0\bar{6}} \approx 0.447$$

Calculation of the Translation:

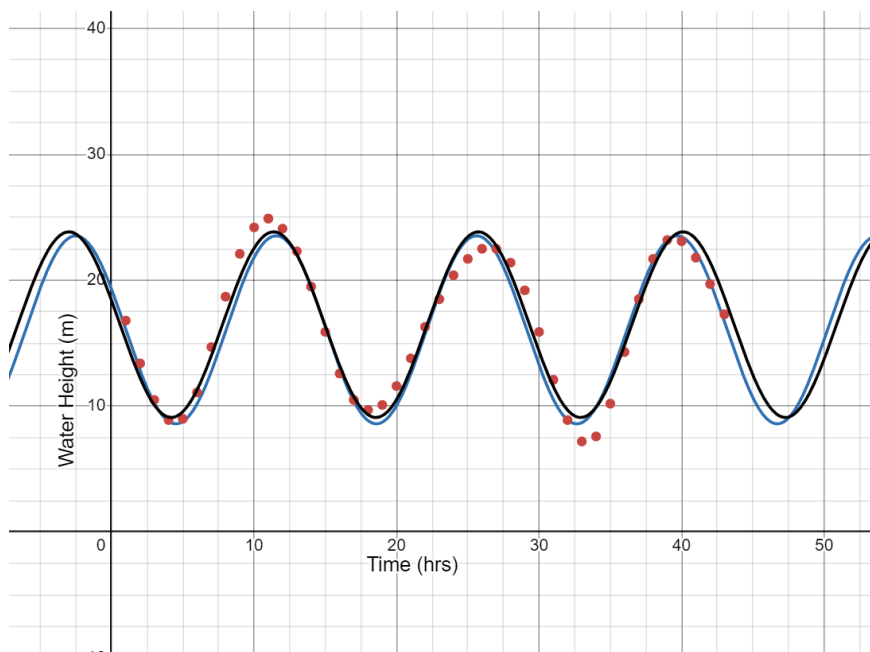
Lastly, we needed to solve for variable h , or the translation. In order to find the translation, we simply used the t value in the starting point of the first period, $(1.2, 16.0\bar{6})$. As a result, our translation was 1.2.

Final Equation:

After solving for every variable using the calculations above, and plugging each one into the equation, $y_1 \sim a \sin(b(t_1 - h)) - k$, our final equation was $h_1(t) = -7.47 \sin(0.447(t - 1.20)) + 16.1$. Following the instructions provided, we named this model $h_1(t)$. The domain of this model is $[1.00, 43.0]$. This is because the equation specifically models tidal data over a 43-hour observation period. Although the sine function can theoretically continue indefinitely in both directions, in this particular context, we are concerned with the specific time frame over which the tidal data is observed.

Desmos will be used to find the model (shown below in Figure 3) along with the corresponding R^2 .

Figure 3: Technologically Generated Line of Best Fit



**Blue line is our line of best fit, whereas the black line is the Desmos line of best fit*

$$h_2(t) = -7.37 \sin(0.437(t_1 - (-6.58)) + 16.5$$

$$R^2 = 0.956$$

Steps to Finding a Regression Model Via Desmos Calculator:

We used the Desmos graphing calculator to generate a regression model for the data provided. We first graphed the data points of the provided data in Desmos, then inserted the regression equation of $y_1 \sim a \sin(b(t_2 - h)) - k$ below. We used the regression equation to find the sine regression model, generating the equation of $y = -7.37 \sin(0.437(t_1 - (-6.58)) + 16.5$. Following the instructions provided, we named this equation $h_2(t)$. Desmos generated us the amplitude (a), the scale factor (b), the horizontal transformation (h), and the principal axis/vertical transformation (k). Hence, this is the technologically generated line of best fit and regression model.

Looking at the graph, we can see that its domain stretches from the start to the end of the portion of the graph shown. We can assume that the domain at least includes the portion Desmos generated. However, mathematically speaking, since the domain of any sine equation is $(-\infty, +\infty)$, it is easy to conclude domain of this model is $(-\infty, +\infty)$. Nonetheless, since this graph is supposed to represent the water height fluctuations over the 43-hour period, the most applicable domain is actually $[1.00, 43.0]$.

Analysis of Both Models (Figures 2 and 3):

Both models appear to fit the data, however, both models do not fit perfectly. The graphed version of the graph we mathematically found by hand shown in Figure 2 strays away from the actual scatterplot, and so does Figure 3. Nonetheless, the black line in Figure 3 is more accurate, hitting more points on the scatterplot compared to Figure 2. Therefore, we think that Figure 3 ($h_2(t)$) is a better model, because it is more accurate.

There are differences because there ought to be some human error in the graph we calculated, and there ought to be some slight differences when calculated by technology. For example, when we calculated the hand drawn graph, we might have measured the points slightly off because the human eye is not as accurate as technology, therefore making our graph slightly less perfect than the one Desmos generated for us.

Application of Sine Model:

Taking the two models we generated and calculated, as explained earlier, we decided that the better model of the two was $h_2(t)$, because it hit the points far more accurately, such as hitting the crux, minimum points of period 2 and 3, and representing the average amplitudes far better. Moreover, we noticed that $h_1(t)$ seemed to be shifted to the right horizontally a little too much, therefore missing some points on the left side of the graph (and the periods on the left side of the graph). Hence, we are going to use $h_2(t)$, which is shown in the equation underneath Figure 3.

Using our model to predict $h(3)$, $h(39)$, $h(85)$, we first briefly looked over the graph on Desmos, lining the t axis up to the points provided (respectively 3, 39, and 85). We made an educated guess on where these points landed on the y axis.

Following these steps, we predicted $h(3)$ to be at $y = 10.0$, $h(39)$ to be at $y = 23.2$, and $h(85)$ to be at $y = 20.0$. To make the process easier, we set the steps for the t axis to be 3 when calculating for $h(3)$, 39 when calculating for $h(39)$, and so on.

Then, to confirm our accuracy, we plugged in the values of 3, 39, and 85 in place of the x value in Desmos. We will round to three significant figures as appropriate.

For $h(3)$, we plugged in 3 in the place of x in Desmos, ending up with the value of 10.0. For $h(39)$, the calculation was 23.4, and for $h(85)$, the calculation was 20.3.

Overall, our predictions were pretty accurate, only missing some by the first decimal point.

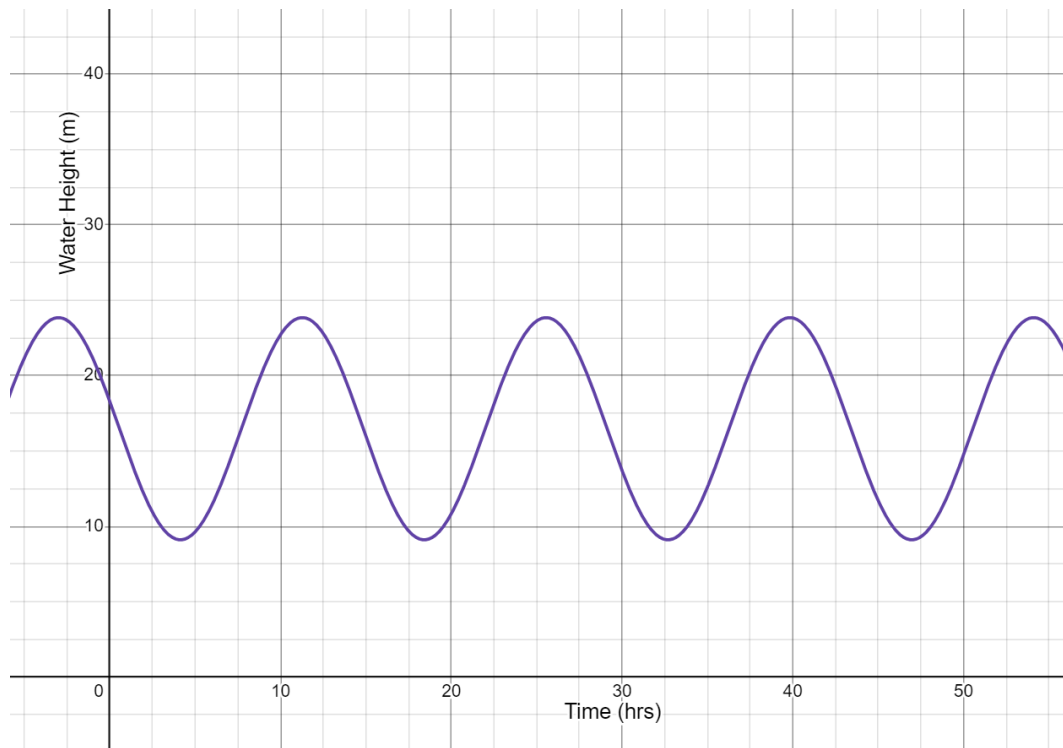
Time Intervals in Sine Model Application:

According to the instructions there needs to be at least 17 meters of water in Victoria Harbour for boats to enter/exit. Here we will determine the time intervals in which the boats can enter or leave the harbor on these two days.

We decided to calculate this graphically, using Desmos. The process will be explained below.

Firstly, we put in the model Desmos calculated as our graph of best fit ($h_2(t)$), as shown below in Figure 4:

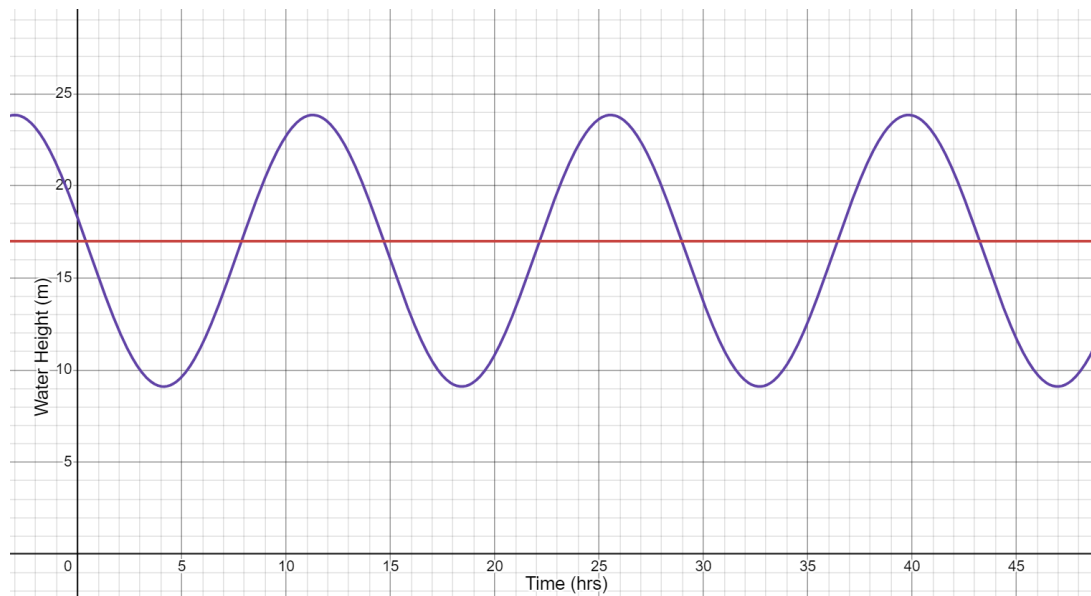
Figure 4: Graph of Best Fit for Intervals



Then, we inputted the equation of $y = 17$ to see which intervals would be above the line, and which intervals would be below. The line will be in red. The intervals above the line would be the time intervals in which boats can enter or leave the harbour on these two days, and the intervals below the line would be the opposite.

The graphical work done in Desmos will be shown in Figure 5 below...

Figure 4: Calculating Time Intervals



Looking at the graph, we found that the t values where the purple line crossed the red line were the following (all values are rounded to three significant figures as appropriate).

$$t = 0.399, 7.86, 14.7, 22.1, 29.0, 36.4, 43.2$$

We noticed that the time intervals of $[1, 7.86) \cup (14.7, 22.1) \cup (29.0, 36.4)$ in the domain are where the water is under the height of 17 meters, whereas the time intervals of $[7.86, 14.7] \cup [22.1, 29.0] \cup [36.4, 43.0]$ were where the water was at least 17 meters or more. Note that we stopped at around $t = 43.0$ and only started the time intervals at $t = 1.00$, as the data given only extended to 43-hours from 1.00 to 43.0, and we are only calculating the intervals between the two days.

Therefore, the time intervals where boats can leave or enter are from around hour 7.86 to 14.7, 22.1 to 29.0, and 36.4 to 43.0.

Summary:

In this paper, we found out the line of best fit through multiple attempted calculations, time intervals where boats can pass in and out of the harbour and analyzed different application scenarios. Moreover, we provided a thorough overview of the steps and processes we went through to find the line of best fit, giving ourselves insight on different calculation techniques (such as using Desmos and hand calculating equations). Through these steps, we learned a lot about application methods, as well as deepened our understandings on sinusoidal functions and real-life applications.

This information is necessary, as waves are crucial to boats coming in and out of the harbour. Additionally, it also showcases the power of nature and the gravitational draw of the moon, where wave heights shift up and down accordingly.

Nonetheless, there were certain limitations with our model. For example, the data given (that was plotted in Figure 1) was not a perfect sine graph, meaning that we even if we generated the best possible line of best fit, it would still not hit all the points. This is a limitation, because it proves that nothing can make the perfect model. Moreover, even though our attempts at creating the best model were excellent, they were lacking a little. For example, when drawing the hand calculated model, we could have more accurately tried to identify the transformations with the human eye. We could have made the steps in the horizontal and vertical axis much smaller values, so that we could have gotten more accurate points for our calculations.